

## PRFM — Prediction Record Formation Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

### Parent Standard:

PRCS — Prediction Record & Continuity Standard

Operational Layer: Prediction Record & Historical Continuity

### Category:

Governance & Enforcement

### Subcategory:

Prediction Governance

Type: Parent Standard Module

### Governed Space:

Prediction Record Formation

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AI-Readable: Yes

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### Protection:

MIP® — Methodological Intellectual Property

### Canonical Language:

English (UCL™)

## Governed Space

PRFM governs how a sufficiently formed Governed Prediction becomes represented as an identifiable, historically anchored, reconstructable Prediction Record capable of preserving its lifecycle from initial governed state through later versioning, historical preservation, status transitions, retirement, and legitimate lifecycle re-entry.

It governs the transition:

Governed Prediction → Initial Historical Prediction State → Governed Prediction Record

Its foundational principle is:

A prediction cannot preserve a history that was never properly recorded at its origin.



## Canonical Definition

The Prediction Record Formation Module (PRFM) defines the operational governance requirements by which a Governed Prediction is established as a persistent Prediction Record with sufficient identity, temporal anchoring, predictive content, methodology references, Prediction Basis references, assumption state, probability and uncertainty state, Prediction Horizon, Predictive Boundary, predictor context, lifecycle status, and provenance references to support subsequent historical continuity and reconstruction.

PRFM creates the historical anchor from which the prediction's governed lifecycle can subsequently be reconstructed.

## Operational Role

PRFM is the first operational module within PRCS.

Its focused operational question is:

What minimum historical state must be established when a Governed Prediction enters the Prediction Record so that its later lifecycle can remain identifiable and reconstructable?

PRFM does not form the prediction itself.

PFOS owns Prediction Formation.

PRFM forms:

the historical record of the prediction that has been formed.

Prediction Formation  $\neq$  Prediction Record Formation

## Internal Module Flow

Prediction Record Formation  $\rightarrow$  Prediction Version Continuity  $\rightarrow$  Prediction History Preservation  $\rightarrow$  Prediction Record State Transition  $\rightarrow$  Prediction Record Re-Entry

### Current Position:

Prediction Record Formation

## 1. Prediction Record Formation

Prediction Record Formation is the governed establishment of the first historically persistent representation of a Governed Prediction.

It converts:

Predictive State

into:

Historically Governable Predictive State

The prediction becomes capable of accumulating a reconstructable lifecycle.

## 2. Prediction vs Prediction Record

A prediction is the predictive claim.

A Prediction Record is the governed historical representation of that claim and its subsequent lifecycle.

Prediction  $\neq$  Prediction Record

The distinction is essential.

## 3. Record Formation Trigger

A Prediction Record should be formed when a prediction reaches a state at which its historical existence becomes materially relevant.

Possible triggers include:

formal prediction formation; publication; communication; operational use; decision reliance; entry into a forecasting system; registration; automated execution eligibility; inclusion in a predictive population.

## 4. Record Formation Timing

The Prediction Record should be formed sufficiently early to prevent later reconstruction from depending solely on memory, retrospective reporting, or outcome knowledge.

Retrospective Reconstruction  $\neq$  Original Record Formation

## 5. Initial Historical State

PRFM establishes the:

Initial Historical Prediction State

This represents the earliest governed state from which historical continuity begins.

## 6. Historical Anchor

The Initial Historical Prediction State becomes the Historical Anchor of the Prediction Record.

Conceptually:

Prediction Origin  $\rightarrow$  Historical Anchor  $\rightarrow$  Prediction Lifecycle

## 7. Prediction Record Identity

Every governed Prediction Record should possess sufficient identity to distinguish it from other Prediction Records.

The implementation may use:

unique identifiers; registry identifiers; persistent references; distributed identifiers; another governed mechanism.

PRFM does not mandate one technical identity system.

## 8. Stable Record Identity

The Prediction Record identity should remain sufficiently stable across later lifecycle changes.

Prediction Change  $\neq$  New Record Identity by Default

## 9. Prediction Identity vs Record Identity

Prediction Identity and Prediction Record Identity may be operationally connected.

They should not be conceptually confused.

The Prediction Object identifies the prediction.

The Prediction Record identifies its governed historical record.

## 10. Record Origin

The Prediction Record should preserve when and under what governed context it originated.

Origin should not be recreated later merely because the record moves to another system.

Migration  $\neq$  New Record Origin

## 11. Origin Timestamp

PRFM should preserve the time at which the Prediction Record entered governed historical existence.

Where relevant, separate timestamps may exist for:

prediction formation; record creation; publication; communication; registration; activation.

These should not be silently collapsed.

## **12. Formation Time vs Record Time**

A prediction may be formed before its formal Prediction Record is created.

Where material:

Prediction Formation Time  $\neq$  Prediction Record Creation Time

Both should remain visible.

## **13. Temporal Precision**

Timestamps should carry sufficient precision for the governance purpose.

Examples:

year; date; hour; minute; second; machine event timestamp.

False precision should not be manufactured.

## **14. Time Source**

Where timing is material, the applicable time source or system context should remain identifiable.

## **15. Prediction Content Snapshot**

PRFM preserves the prediction content as it existed at record formation.

This becomes the first historical content snapshot.

## **16. Content Fidelity**

The recorded prediction should faithfully represent what was actually predicted.

It should not be retrospectively improved for clarity after the outcome becomes known.

## **17. Prediction Object Reference**

The record should identify the Prediction Object governed under POPS.

This preserves what exactly was being predicted.

## **18. Predictive Purpose Reference**

Where material, the record should preserve the predictive purpose defined under POPS.

The same predictive statement used for different purposes may carry different governance implications.

## **19. Prediction Horizon Reference**

The applicable Prediction Horizon should be recorded.

A later extension of the horizon must not alter the original record.

## **20. Predictive Boundary Reference**

Material Predictive Boundaries should remain associated with the Initial Historical Prediction State.

Examples:

geography; population; market; technology; product; event; scenario.

## 21. Methodology Reference

The Prediction Record should identify the applicable PMES-governed Prediction Methodology or methodology version where available.

## 22. Methodology State at Formation

PRFM preserves the methodology state that actually applied when the prediction was formed.

A future methodology upgrade does not alter that historical state.

Future Methodology  $\neq$  Historical Methodology

## 23. Prediction Basis Reference

The applicable PBAS-governed Prediction Basis should remain linkable to the Initial Historical Prediction State.

## 24. Basis Time Context

The record should preserve when relevant Prediction Basis information was available.

This helps prevent later information from being retrospectively attributed to the original prediction.

## 25. Assumption State

Material PAS-governed assumptions applicable at formation should remain identifiable.

## 26. Assumption Snapshot

Where assumptions are versioned separately, PRFM may preserve a reference to the applicable Assumption Snapshot rather than duplicating the entire assumption structure.

## 27. Formation Reference

The record should preserve the PFOS-governed Prediction Formation state or applicable reference.

PRFM does not reproduce PFOS mechanics.

## 28. Probability State

Where probability is present, the original probability should remain historically recorded.

Example:

Probability at Record Formation: 42%

This value must not later be replaced by:

Final Probability: 91%

## 29. Uncertainty State

Applicable uncertainty representation should remain associated with the initial prediction state.

## 30. Confidence State

Where confidence is separately represented, it should remain distinguishable from probability.

Probability  $\neq$  Confidence

## 31. Range State

Where the prediction contains a range, the original range should remain recorded.

Example:

€110–125 million

A later narrowing of the range creates a later state.

## **32. Conditionality State**

Conditional predictions should preserve their original conditions.

Example:

If Regulation X is adopted by June, commercial deployment is expected by Q4.

The condition is part of the historical predictive meaning.

## **33. Scenario State**

Scenario-specific predictions should preserve the scenario under which they were formed.

## **34. Feasibility Reference**

Where PFES feasibility governance materially applies, the applicable feasibility state may be referenced.

## **35. Collective Prediction Reference**

Where the prediction originates from a collective mechanism, relevant CPGS state should remain identifiable.

## **36. Predictor Reference**

The record should preserve sufficient reference to the human, model, agent, organization, collective, or predictive mechanism that produced the prediction where applicable.

## **37. Predictor Type**

Possible Predictor Types may include:

Human

AI Model

Autonomous Agent

Organization

Expert Group

Collective Prediction

Prediction Market

Hybrid Human-AI System

## **38. Predictor Version**

Where a model, agent, or system has versions, the applicable version should remain historically attributable.

## **39. Predictor Identity Boundary**

PRFM records the predictor reference relevant to the prediction.

It does not establish the broader legal or digital identity of that predictor.

## 40. Information Condition

Where materially available, the Initial Prediction Record may preserve relevant information-condition context.

This may later support Predictor–Reality Proximity analysis.

## 41. Public-Information Context

A prediction based solely on publicly available information may preserve that context where governed.

## 42. Information-Advantage Context

Where known and legitimate to record, unusual informational advantage may be preserved.

PRFM does not infer misconduct.

## 43. Outcome Knowledge Boundary

The Initial Historical Prediction State should be established before later outcome knowledge can silently contaminate the record.

## 44. Record Formation After Outcome

A Prediction Record created only after the outcome is known should be clearly classified as retrospective where applicable.

It must not automatically be treated as equivalent to a contemporaneous historical record.

Retrospective Record  $\neq$  Contemporaneous Prediction Record

## 45. Contemporaneous Record

A Contemporaneous Prediction Record is formed sufficiently close to the predictive event to provide stronger historical anchoring.

## 46. Retrospective Record

A Retrospective Prediction Record may still have value.

But its provenance and evidentiary limitations should remain visible.

## 47. Record Formation Source

PRFM should preserve the source from which the Initial Prediction Record was formed where material.

Examples:

forecasting platform; signed report; database; model output; human submission; prediction market; API; registered predictive record.

## 48. Source Preservation

Source reference helps establish how the recorded predictive state relates to the originating representation.

## 49. Source vs Record

The source may disappear.

The governed Prediction Record should preserve enough reference to reconstruct the historical relationship where appropriate.

Source  $\neq$  Prediction Record

## 50. Communication State at Formation

A prediction may initially be:

Private

Internal

Restricted

Public

Communicated to Specific Recipients

Where material, this state should remain identifiable.

## 51. Reliance State at Formation

The prediction may initially be:

Not Relied Upon

Reliance Eligible

Already Relied Upon

where PCRS-governed context exists.

## 52. Lifecycle Status

PRFM establishes the initial lifecycle status of the Prediction Record.

Possible states may include:

Formed

Active

Registered

Communicated

Operational

Restricted

depending on the implementation.

## 53. Record Completeness

Initial Record Completeness determines whether sufficient historical information exists to support future reconstruction.

## 54. Required vs Optional Record Elements

Not every prediction requires identical record depth.

PRFM distinguishes:

Required Historical Elements

from:

Context-Dependent Historical Elements

## 55. Minimum Historical Sufficiency

At minimum, a useful Prediction Record should normally make it possible to identify:

what was predicted; which prediction it was; when the record was formed; the applicable historical state; the Prediction Horizon where material; the predictor or source where material; relevant probability or uncertainty where present.

Additional elements depend on context.

## 56. Record Sufficiency State

PRFM may classify initial record sufficiency as:

Sufficient

Sufficient with Qualification

Partially Sufficient

Material Information Missing

Insufficient

Unable to Determine

## 57. Missing Historical Element

Where a material element is missing, PRFM should identify the gap rather than invent the missing value.

## 58. Unknown ≠ Absent

If a historical element cannot be determined, the record should distinguish:

Unknown

from:

Not Applicable

and:

Absent

## 59. Record Qualification

A Prediction Record may remain usable with qualifications.

Example:

Original timestamp known only to calendar date

rather than exact minute.

## 60. Record Integrity at Formation

The Initial Historical Prediction State should be protected from silent modification after it becomes historical.

## 61. Historical Freeze

Historical Freeze means that the initial state becomes an identifiable historical state once subsequent material change occurs.

It does not prohibit governed correction.

## 62. Correction Mechanism

If an Initial Prediction Record contains an error, correction should preserve:

Original Record → Correction Event → Corrected Representation

where material.

### **63. Correction ≠ Replacement**

The original historical state should remain reconstructable where governance requires it.

### **64. Clarification**

A clarification may explain ambiguous wording without changing predictive meaning.

The clarification should remain distinguishable from the original state.

### **65. Material Correction**

A correction that changes predictive meaning should normally create a new historical state rather than silently modifying the original.

### **66. Record Provenance**

PRFM should preserve sufficient provenance to answer:

Where did this Prediction Record come from?

### **67. Provenance Elements**

Relevant provenance may include:

originating system; originating predictor; record creator; timestamp; source reference; import mechanism; transformation reference.

### **68. Provenance Boundary**

PRFM preserves prediction-record provenance sufficient for PRCS continuity.

It does not replace comprehensive provenance governance.

### **69. Human-Created Record**

Where a human manually records a prediction, the distinction between:

Predictor

and:

Record Creator

may be material.

### **70. Machine-Created Record**

Where a system automatically records model output, the system and model references should remain identifiable where material.

### **71. Imported Record**

A prediction imported from another system should preserve its prior origin where available.

Import time must not become false prediction origin.

### **72. Migration Boundary**

Moving an existing Prediction Record into a new system is not Record Formation of a new prediction.

Import ≠ New Prediction Origin

## 73. Duplicate Record

The same prediction may accidentally or intentionally appear in multiple records.

PRFM should support identification of possible duplicates.

## 74. Duplicate ≠ Independent Prediction

A duplicate copy should not automatically increase prediction count in later performance analysis.

## 75. Record Linkage

Where multiple representations refer to the same Prediction Record, they may be linked through a persistent record relationship.

## 76. Canonical Record

An implementation may designate one representation as the Canonical Record.

PRFM does not mandate centralized storage.

## 77. Distributed Record

Prediction history may exist across multiple systems.

PRFM should preserve enough identity and provenance for later continuity governance.

## 78. Record Ownership Boundary

Creation of a Prediction Record does not itself establish intellectual-property or legal ownership.

## 79. Confidential Prediction Record

Private or confidential predictions may still require governed Record Formation.

Private Prediction ≠ History-Free Prediction

## 80. Public Prediction Record

Public publication may strengthen temporal evidence but does not replace governance of record identity, methodology, basis, or context.

## 81. Prediction Market Record Formation

For prediction markets, the record may preserve:

market identity; market question; resolution rule; probability snapshot; timestamp; market state.

A market price alone is not the complete Prediction Record.

## 82. AI Prediction Record Formation

For AI-generated predictions, the record may preserve:

model identity; model version; relevant prompt or task reference where governed; methodology reference; output; probability; timestamp; system context.

## 83. Autonomous Agent Record Formation

Where an autonomous agent forms or uses predictions, PRFM supports machine-generated historical anchoring before the prediction influences action.

## 84. Human-AI Prediction Record

Where humans and AI jointly form a prediction, the record may preserve the applicable Human-AI formation structure.

## 85. Prediction Record Schema

Implementations may use a structured Prediction Record Schema.

PRFM governs semantic requirements rather than prescribing one database schema.

## 86. Machine Readability

Prediction Record Formation should support machine-readable representation where appropriate.

## 87. Machine Reconstruction

A machine-readable Initial Prediction Record should make it possible to determine:

what was predicted

when

under which state

by whom or what

with what probability or uncertainty where applicable

under which horizon and boundary

## 88. Record Validation

Implementations may validate whether required Record Formation fields are present and internally consistent.

VALIDOS™ may govern broader validation structures.

## 89. Record Integrity Check

PRFM may support checks for:

missing identity; missing timestamp; impossible timestamp ordering; missing Prediction Object; malformed probability; missing source; duplicate record; inconsistent horizon.

## 90. Record Formation Failure

Where sufficient initial history cannot be established, the system should not manufacture a complete Prediction Record.

Possible result:

Prediction Record Formation Incomplete

## 91. Governance Requirements

A governed PRFM implementation should be able to:

1. determine when Prediction Record Formation is required; 2. establish an Initial Historical Prediction State; 3. create a Historical Anchor;

4. establish Prediction Record Identity; 5. preserve stable identity across later changes; 6. distinguish Prediction Identity from Record Identity; 7. preserve Record Origin; 8. preserve Origin Timestamp; 9. distinguish Prediction Formation Time from Record Creation Time; 10. preserve appropriate temporal precision; 11. preserve applicable Time Source; 12. preserve Prediction Content Snapshot; 13. maintain Content Fidelity; 14. preserve Prediction Object reference; 15. preserve Predictive Purpose reference where

material; 16. preserve Prediction Horizon; 17. preserve Predictive Boundary; 18. preserve Prediction Methodology reference; 19. preserve Methodology State at Formation; 20. preserve Prediction Basis reference; 21. preserve Basis Time Context; 22. preserve Assumption State; 23. preserve Formation reference; 24. preserve Probability State; 25. preserve Uncertainty State; 26. preserve Confidence State where applicable; 27. preserve Range State; 28. preserve Conditionality State; 29. preserve Scenario State; 30. preserve Feasibility reference where applicable; 31. preserve Collective Prediction reference where applicable; 32. preserve Predictor Reference; 33. preserve Predictor Type; 34. preserve Predictor Version where applicable; 35. preserve Predictor Identity Boundary; 36. preserve Information Condition where material; 37. preserve Outcome Knowledge Boundary; 38. identify retrospective Record Formation; 39. distinguish Retrospective from Contemporaneous Records; 40. preserve Record Formation Source; 41. preserve Source Reference; 42. distinguish Source from Prediction Record; 43. preserve Communication State where applicable; 44. preserve Reliance State where applicable; 45. establish initial Lifecycle Status; 46. determine Record Completeness; 47. distinguish Required from Optional Record Elements; 48. determine Minimum Historical Sufficiency; 49. classify Record Sufficiency; 50. identify Missing Historical Elements; 51. distinguish Unknown, Absent, and Not Applicable; 52. preserve Record Qualification; 53. preserve Record Integrity at Formation; 54. support Historical Freeze; 55. support governed Correction; 56. distinguish Correction from Replacement; 57. preserve Clarification; 58. identify Material Correction; 59. preserve Record Provenance; 60. preserve applicable Provenance Elements; 61. distinguish Predictor from Record Creator; 62. support Machine-Created Records; 63. support Imported Records; 64. prevent Import from creating false origin; 65. identify Duplicate Records; 66. prevent duplicates from becoming independent predictions by default; 67. support Record Linkage; 68. support Canonical Record references where applicable;

69. support Distributed Records; 70. preserve ownership boundary; 71. support Confidential Prediction Records; 72. support Public Prediction Records; 73. support Prediction Market Record Formation; 74. support AI Prediction Record Formation; 75. support Autonomous Agent Record Formation; 76. support Human-AI Prediction Records; 77. support structured Prediction Record Schemas; 78. support Machine Readability; 79. support Machine Reconstruction; 80. support Record Validation; 81. support Record Integrity Checks; 82. identify Record Formation Failure; 83. preserve formation history; 84. preserve record traceability; 85. produce a Governed Initial Prediction Record.

## 92. Governance Logic

PRFM applies the following operational logic:

Has a sufficiently formed Governed Prediction reached a state requiring historical preservation?

If no:

→ Prediction Record Formation is not yet required.

If yes:

→ establish Prediction Record Identity.

Is the original predictive state sufficiently identifiable?

If no:

→ classify Record Formation as incomplete or retrospective.

If yes:

→ preserve Prediction Content Snapshot.

Can formation time and record creation time be established?

If yes:

→ preserve both where materially different.

If no:

→ preserve temporal qualification.

Are Prediction Object, Horizon, Boundary, Methodology, Basis, Assumptions, Probability, Uncertainty, and Predictor Context available where applicable?

→ preserve available governed references.

Are material elements missing?

If yes:

→ identify Historical Gaps.

Do not invent values.

Was the record created after outcome knowledge became available?

If yes:

→ classify as retrospective where appropriate.

Was the record imported from another system?

If yes:

→ preserve original origin and lineage.

Do not create false new origin.

Does a potentially identical Prediction Record already exist?

If yes:

→ evaluate duplication and linkage.

Is sufficient historical state available?

If yes:

→ establish:

Governed Initial Prediction Record

If partially:

→ establish qualified record state.

Then:

→ freeze the initial historical state against silent replacement;

→ enable subsequent version continuity.

Canonical transition:

Governed Prediction → Historical Anchoring → Initial Prediction Record → Version-Ready Prediction Record

## 93. Minimum Implementation Framework

A. Prediction Record ID

Persistent identifier for the governed Prediction Record.

B. Prediction Reference

Reference to the underlying Governed Prediction.

C. Initial Historical State

First governed historical representation of the prediction.

#### D. Record Origin

Originating predictive and record context.

#### E. Temporal Anchor

Prediction formation time, record creation time, and applicable time qualification.

#### F. Prediction Content Snapshot

Exact or sufficiently faithful representation of the prediction at record formation.

#### G. Prediction Object & Purpose Reference

Applicable POPS-governed object and purpose.

#### H. Horizon & Boundary Reference

Applicable Prediction Horizon and Predictive Boundary.

#### I. Methodology Reference

Applicable PMES methodology and version where available.

#### J. Basis & Assumption Reference

Applicable PBAS Prediction Basis and PAS assumption state.

#### K. Probability & Uncertainty State

Applicable PPUS probability, confidence, range, uncertainty, and conditionality state.

#### L. Predictor & Source Reference

Applicable predictor, model, agent, collective, platform, or source context.

#### M. Initial Lifecycle Status

Prediction status when the historical record is established.

#### N. Record Sufficiency & Qualification

Completeness state, missing elements, retrospective status, and other material qualifications.

#### O. Provenance & Integrity Record

Origin, creator, source, timestamp, correction state, duplicate relationship, and integrity information necessary for later continuity.

## 94. Governance Outputs

PRFM may produce:

Governed Initial Prediction Record Prediction Record ID Initial Historical Prediction State Historical Anchor  
Record Origin Temporal Anchor Prediction Content Snapshot Prediction Object Reference Predictive  
Purpose Reference Prediction Horizon Reference

Predictive Boundary Reference Methodology Reference Prediction Basis Reference Assumption State  
Reference Probability State Uncertainty State Predictor Reference Predictor Type Predictor Version  
Information Condition Reference Record Formation Source Initial Lifecycle Status Record Sufficiency State  
Historical Gap Record Retrospective Record Qualification Record Provenance Record Integrity State  
Duplicate Record Indicator Record Linkage Reference Formation Traceability Record

The primary output is:

Governed Initial Prediction Record

This becomes the historical anchor for the remaining PRCS lifecycle.

## 95. Critical Distinctions

Prediction  $\neq$  Prediction Record

Prediction Formation  $\neq$  Prediction Record Formation

Retrospective Reconstruction  $\neq$  Original Record Formation

Prediction Change  $\neq$  New Record Identity by Default

Migration  $\neq$  New Record Origin

Prediction Formation Time  $\neq$  Prediction Record Creation Time

Future Methodology  $\neq$  Historical Methodology

Probability  $\neq$  Confidence

Retrospective Record  $\neq$  Contemporaneous Prediction Record

Source  $\neq$  Prediction Record

Unknown  $\neq$  Absent

Correction  $\neq$  Replacement

Import  $\neq$  New Prediction Origin

Duplicate  $\neq$  Independent Prediction

Private Prediction  $\neq$  History-Free Prediction

Public Publication  $\neq$  Complete Prediction Governance

Record Formation  $\neq$  Proof of Predictive Quality

Historical Record  $\neq$  Prediction Validation

## 96. Enterprise Application

PRFM can govern Prediction Record Formation across:

AI forecasting systems; enterprise forecasting; financial forecasting; investment analysis; insurance; prediction markets; industrial prediction; predictive maintenance; supply-chain forecasting; energy forecasting; infrastructure; technology forecasting; public-sector forecasting; expert forecasting; Human-AI forecasting; model ensembles; autonomous agents; multi-agent systems.

Its function remains constant:

establish the prediction as a historically anchored governed object before later change makes its original state difficult or impossible to reconstruct.

## 97. Use Case 1 — A Company Claims It Predicted a Market Collapse Before It Happened

An investment firm publishes a report after a major market collapse stating:

“Our forecasting system predicted this event with 82% probability.”

The firm presents a database entry showing:

82% probability

but the entry was created after the market collapse.

Earlier internal documents contain several different forecasts, but it is unclear which one represented the actual prediction before the outcome became known.

The critical governance question is:

Did a contemporaneous Governed Prediction Record containing the 82% probability actually exist before the outcome, or was the record reconstructed retrospectively?

PRFM examines:

Prediction Record origin; Prediction Formation Time; Record Creation Time; source reference; probability state; methodology version; Prediction Basis timing; predictor identity; contemporaneous evidence.

If the record was formed after the outcome:

Retrospective Prediction Record

may be the legitimate classification.

The prediction may still have existed earlier.

But the later record must not automatically be treated as equivalent to a contemporaneous historical anchor.

Retrospective Record  $\neq$  Contemporaneous Prediction Record

Outcome Knowledge  $\neq$  Permission to Manufacture Prediction History

## **98. Use Case 2 — An AI Model Is Upgraded and the Original Prediction Context Disappears**

An enterprise AI system predicts:

65% probability of a critical supply disruption within 90 days.

The prediction is used in operational planning.

Two weeks later, the company upgrades:

Model v4.1  $\rightarrow$  Model v5.0

The new system recalculates the event at:

38%

and replaces the previous value in the operational database.

Months later, management asks:

“What prediction was actually available when the supply decision was made?”

The database now contains only:

38% — Model v5.0

Without PRFM, the original predictive state may no longer be reconstructable.

A governed PRFM record would have preserved:

Prediction Record ID

Prediction: Supply disruption within 90 days

Probability: 65%

Model: v4.1

Prediction Horizon: 90 days

Prediction Formation Time

Record Creation Time

Methodology Reference

Prediction Basis Reference

Initial Lifecycle State

When Model v5.0 later creates a revised predictive state, the original record remains historically anchored.

The critical distinction is:

New Model Output  $\neq$  Replacement of Historical Prediction State

The later PRCS modules can then preserve:

v4.1 Prediction State  $\rightarrow$  v5.0 Prediction State  $\rightarrow$  Subsequent History

rather than leaving only the latest database value.

## 99. Canonical Boundary

PRFM governs the formation of the initial historically persistent Prediction Record from a sufficiently formed Governed Prediction. It establishes the Historical Anchor, Prediction Record Identity, temporal origin, Prediction Content Snapshot, applicable methodology, Prediction Basis, assumption, probability, uncertainty, horizon, boundary, predictor, source, lifecycle, provenance, sufficiency, and integrity context necessary for subsequent historical continuity. It does not form the underlying prediction, determine predictive methodology, validate predictive quality, determine Operational Reality, assess Predictive Performance, establish legal ownership, or govern later version transitions, complete history preservation, record-state transitions, or lifecycle re-entry.

PFOS owns:

Prediction Formation

PRFM owns:

Prediction Record Formation

PVCM subsequently owns:

Prediction Version Continuity

PHPM subsequently owns:

Prediction History Preservation

PRSTM subsequently owns:

Prediction Record State Transition

PRRM subsequently owns:

Prediction Record Re-Entry

## 100. Canonical Closing Statement

Prediction history does not begin when someone later asks what was predicted. It begins when the prediction first becomes historically identifiable.

PRFM creates that anchor before revisions, new information, model upgrades, operational use, or outcome knowledge can blur the original predictive state.

A prediction can change a thousand times after formation. Governance must still be able to show exactly where its history began.

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